Trabajo Encargado de Cálculo Integral 2da Unidad

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I Halle la fórmula para cada una de las siguientes sumatorias

1. $\sum_{i=1}^{n} i^4$

Usamos la fórmula de Faulhaber para $\sum i^4$:

$$\sum_{i=1}^{n} i^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$\boxed{\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}}$$

2. $\sum_{i=1}^{n} i^5$

Usamos la fórmula de Faulhaber para $\sum i^5$:

$$\sum_{i=1}^{n} i^5 = \frac{n^2(n+1)^2(2n^2+2n-1)}{12}$$

$$\boxed{\frac{n^2(n+1)^2(2n^2+2n-1)}{12}}$$

3. $\sum_{i=1}^{n} \left(\sqrt{9+x}\right)^i$

Es una progresión geométrica con razón $r = \sqrt{9+x}$.

$$\text{La suma es: } S_n = a_1 \frac{r^n - 1}{r - 1}$$

$$\text{Donde } a_1 = \sqrt{9+x}$$

$$S_n = \sqrt{9+x} \cdot \frac{(\sqrt{9+x})^n - 1}{\sqrt{9+x} - 1}$$

$$\boxed{\frac{\sqrt{9+x}\left[(\sqrt{9+x})^n - 1\right]}{\sqrt{9+x} - 1}}$$

4. $\displaystyle\sum_{i=1}^{n} \frac{1}{16i^2-1}$

$$\begin{aligned}
\text{Factorizamos el denominador: } 16i^2-1 &= (4i-1)(4i+1)\\
\text{Fracciones parciales: } \frac{1}{(4i-1)(4i+1)} &= \frac{A}{4i-1}+\frac{B}{4i+1}\\
1 &= A(4i+1)+B(4i-1)\\
\text{Si } i=1/4 \implies 1 &= 2A \implies A=1/2\\
\text{Si } i=-1/4 \implies 1 &= -2B \implies B=-1/2\\
\text{La suma es: } \sum_{i=1}^{n}\frac{1}{2}\left(\frac{1}{4i-1}-\frac{1}{4i+1}\right) &= \frac{1}{2}\left[\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{9}\right)+\cdots+\left(\frac{1}{4n-1}-\frac{1}{4n+1}\right)\right]
\end{aligned}$$

Es telescópica, se cancelan términos intermedios:

$$= \frac{1}{2}\left(\frac{1}{3}-\frac{1}{4n+1}\right)$$

$$\boxed{\frac{1}{2}\left(\frac{1}{3}-\frac{1}{4n+1}\right)=\frac{2n-1}{3(4n+1)}}$$

5. $\displaystyle\sum_{i=1}^{n} \tan(aix)$

Esta sumatoria no posee una fórmula cerrada elemental conocida en términos de funciones trigonométricas básicas para un n arbitrario. Se deja indicada la suma.

$$\boxed{\sum_{i=1}^{n}\tan(aix)}$$

6. $\displaystyle\sum_{i=1}^{n} sen^i(ax)$

Es una progresión geométrica con razón $r=\sin(ax)$.

$$\begin{aligned}
\text{La suma es: } S_n &= a_1\frac{r^n-1}{r-1}\\
\text{Donde } a_1 &= \sin(ax)\\
S_n &= \sin(ax)\frac{\sin^n(ax)-1}{\sin(ax)-1}
\end{aligned}$$

$$\boxed{\frac{\sin(ax)(\sin^n(ax)-1)}{\sin(ax)-1}}$$

II Encontrar el área de la región exacta, expresar como límite de sumas de Riemann, con particiones iguales

7. Hallar el área de la región acotada por $y = 5x - 6$, el eje x y las rectas $x = 1$ y $x = 5$.

$$\text{Intervalo: } [1,5]. \quad \Delta x = \frac{5-1}{n} = \frac{4}{n}. \quad x_i = 1 + \frac{4i}{n}.$$

$$\text{Suma de Riemann: } A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$$

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} \left(5\left(1 + \frac{4i}{n}\right) - 6\right)\frac{4}{n}$$

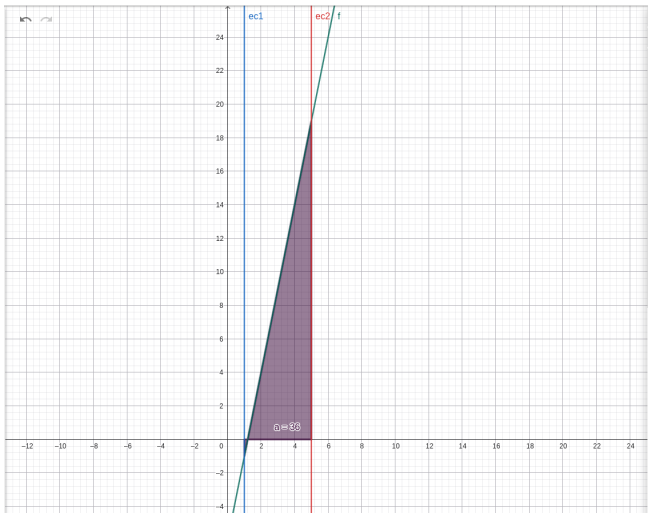
Evaluando la integral definida:

$$A = \int_1^5 (5x-6)\,dx = \left[\frac{5x^2}{2} - 6x\right]_1^5$$

$$= \left(\frac{125}{2} - 30\right) - \left(\frac{5}{2} - 6\right)$$

$$= \frac{65}{2} - \left(-\frac{7}{2}\right) = \frac{72}{2} = 36$$

$$\boxed{36}$$



8. Hallar el área de la región acotada por $y = 5\sqrt{x}$, el eje x y las rectas $x = 1$ y $x = 4$.

$$\text{Intervalo: } [1,4]. \quad \Delta x = \frac{3}{n}. \quad x_i = 1 + \frac{3i}{n}.$$

$$\text{Suma de Riemann: } A = \lim_{n\to\infty} \sum_{i=1}^{n} 5\sqrt{1 + \frac{3i}{n}} \cdot \frac{3}{n}$$

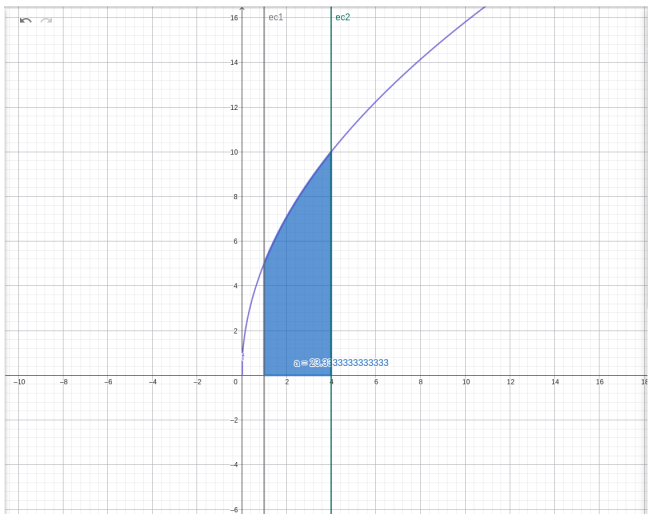
Evaluando la integral:

$$A = \int_1^4 5\sqrt{x}\,dx = 5\int_1^4 x^{1/2}\,dx$$

$$= 5\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{10}{3}(4^{3/2} - 1^{3/2})$$

$$= \frac{10}{3}(8-1) = \frac{70}{3}$$

$$\boxed{\frac{70}{3}}$$



9. Hallar el área de la región acotada por $f(x) = 16 - x^2$, el eje x y las rectas $x = 1$ y $x = 4$.

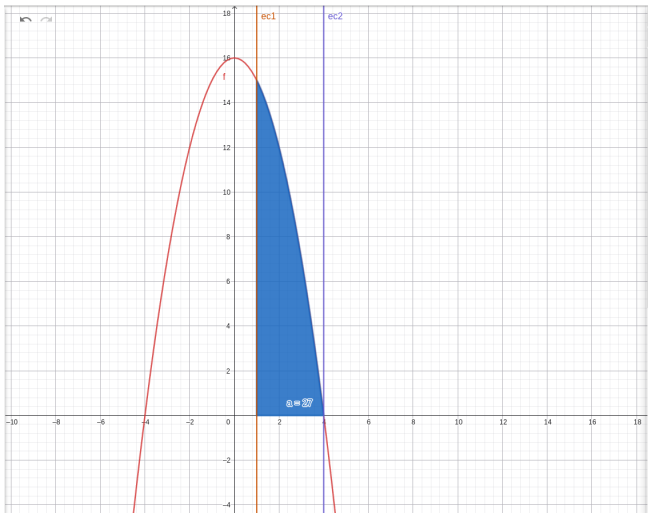
$$\text{Intervalo: } [1,4]. \quad \Delta x = \frac{3}{n}. \quad x_i = 1 + \frac{3i}{n}.$$

$$\text{Suma de Riemann: } A = \lim_{n\to\infty} \sum_{i=1}^{n} \left(16 - \left(1 + \frac{3i}{n}\right)^2\right) \frac{3}{n}$$

Evaluando la integral:

$$\begin{aligned} A = \int_1^4 (16 - x^2)\, dx &= \left[16x - \frac{x^3}{3}\right]_1^4 \\ &= \left(64 - \frac{64}{3}\right) - \left(16 - \frac{1}{3}\right) \\ &= \frac{128}{3} - \frac{47}{3} = \frac{81}{3} = 27 \end{aligned}$$

$$\boxed{27}$$



10. Hallar el área de la región acotada por $y = 5 - |x|$, el eje x y las rectas $x = -5$ y $x = 5$.

La función es par y simétrica respecto al eje y.

Intervalo: $[-5, 5]$. Podemos calcular $2\int_0^5 (5 - x)\,dx$.

Suma de Riemann (para $[0, 5]$): $\Delta x = \dfrac{5}{n}, x_i = \dfrac{5i}{n}$.

$$A = 2\lim_{n\to\infty}\sum_{i=1}^{n}\left(5 - \frac{5i}{n}\right)\frac{5}{n}$$

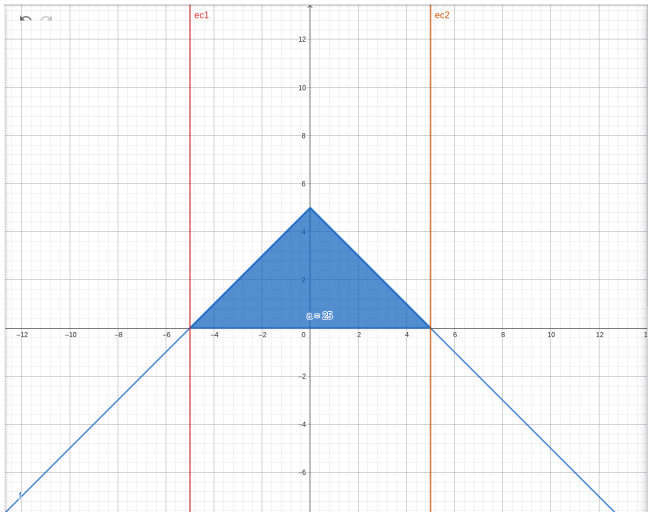
Evaluando geométricamente (triángulo base 10, altura 5):

$$A = \frac{1}{2}\cdot 10\cdot 5 = 25$$

O por integral: $\displaystyle\int_{-5}^{5}(5 - |x|)\,dx = 2\left[5x - \frac{x^2}{2}\right]_0^5$

$$= 2\left(25 - \frac{25}{2}\right) = 2\left(\frac{25}{2}\right) = 25$$

$$\boxed{25}$$



11. Hallar el área de la región acotada por $y = \tan x$, el eje x y las rectas $x = 0$ y $x = \frac{\pi}{3}$.

Intervalo: $[0, \pi/3]$. $\Delta x = \dfrac{\pi}{3n}$. $x_i = \dfrac{i\pi}{3n}$.

Suma de Riemann: $A = \displaystyle\lim_{n\to\infty}\sum_{i=1}^{n}\tan\left(\frac{i\pi}{3n}\right)\frac{\pi}{3n}$

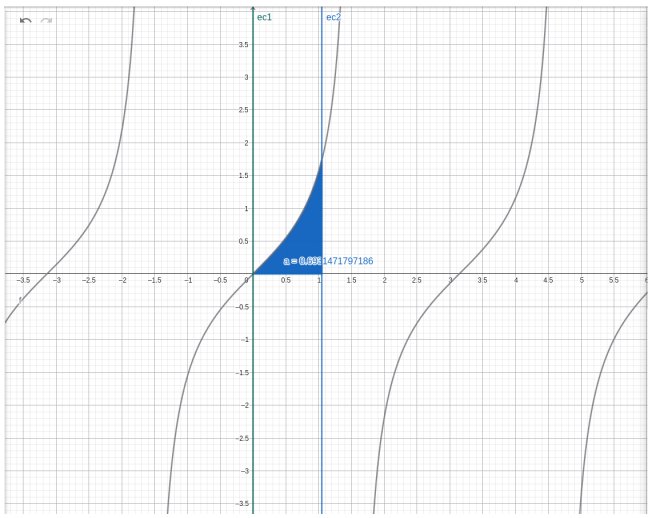
Evaluando la integral:

$$A = \int_0^{\pi/3}\tan x\,dx = [\ln|\sec x|]_0^{\pi/3}$$

$$= \ln(\sec(\pi/3)) - \ln(\sec(0))$$

$$= \ln(2) - \ln(1) = \ln 2$$

$$\boxed{\ln 2}$$

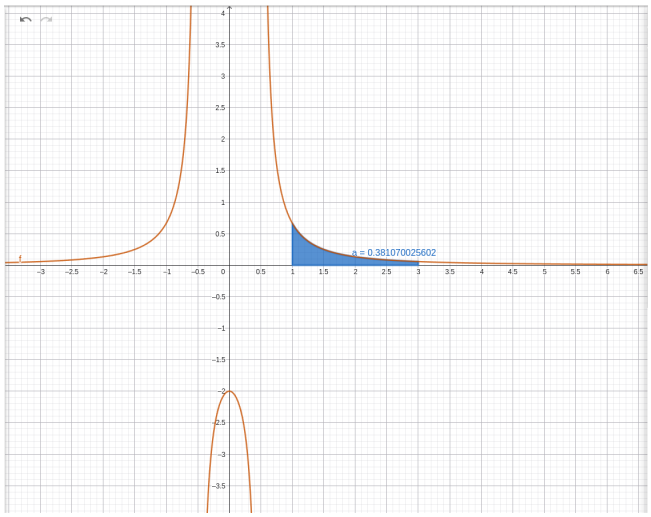


III Halle las siguientes integrales definidas

12. $\displaystyle\int_1^3 \left(\frac{1}{2x-1} - \frac{1}{2x+1}\right) dx$

$$
\begin{aligned}
I &= \int_1^3 \frac{dx}{2x-1} - \int_1^3 \frac{dx}{2x+1} \\
&= \left[\frac{1}{2}\ln|2x-1|\right]_1^3 - \left[\frac{1}{2}\ln|2x+1|\right]_1^3 \\
&= \frac{1}{2}(\ln 5 - \ln 1) - \frac{1}{2}(\ln 7 - \ln 3) \\
&= \frac{1}{2}\ln 5 - \frac{1}{2}\ln\frac{7}{3} = \frac{1}{2}\ln\left(\frac{5}{7/3}\right) \\
&= \frac{1}{2}\ln\frac{15}{7}
\end{aligned}
$$

$$\boxed{\frac{1}{2}\ln\frac{15}{7}}$$



13. $\displaystyle\int_2^4 \frac{e^{\ln x}}{x^2+7}\,dx$

Simplificamos el integrando: $e^{\ln x} = x$.

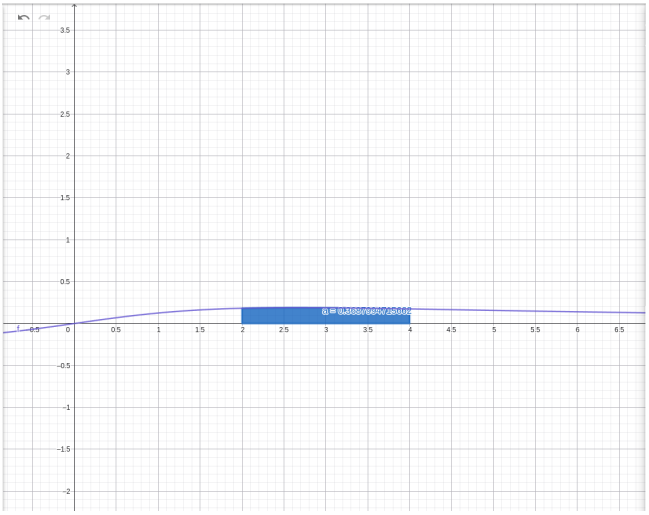
$$I = \int_2^4 \frac{x}{x^2+7}\,dx$$

Sustitución: $u = x^2+7 \implies du = 2x\,dx \implies x\,dx = du/2$.

Límites: $x = 2 \to u = 11; \quad x = 4 \to u = 23$.

$$I = \int_{11}^{23} \frac{1}{u}\frac{du}{2} = \frac{1}{2}[\ln|u|]_{11}^{23}$$
$$= \frac{1}{2}(\ln 23 - \ln 11) = \frac{1}{2}\ln\frac{23}{11}$$

$$\boxed{\frac{1}{2}\ln\frac{23}{11}}$$



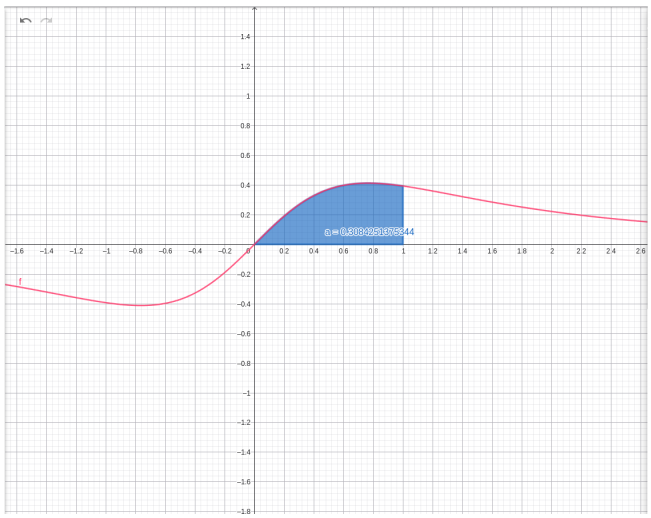
14. $\displaystyle\int_0^1 \frac{arctgx}{x^2+1}\,dx$

Sustitución: $u = \arctan x \implies du = \dfrac{dx}{x^2+1}$.

Límites: $x = 0 \to u = 0; \quad x = 1 \to u = \pi/4$.

$$I = \int_0^{\pi/4} u\,du = \left[\frac{u^2}{2}\right]_0^{\pi/4}$$
$$= \frac{1}{2}\left(\frac{\pi}{4}\right)^2 - 0 = \frac{\pi^2}{32}$$

$$\boxed{\frac{\pi^2}{32}}$$



15. $\displaystyle\int_6^9 \frac{3\ln x - 5}{x}\,dx$

$$\begin{aligned} I &= \int_6^9 \left(3\frac{\ln x}{x} - \frac{5}{x}\right) dx \\ &= 3\int_6^9 \frac{\ln x}{x} dx - 5\int_6^9 \frac{dx}{x} \end{aligned}$$

Para la primera integral: $u = \ln x, du = dx/x$.

$$\begin{aligned} I &= 3\left[\frac{(\ln x)^2}{2}\right]_6^9 - 5\left[\ln|x|\right]_6^9 \\ &= \frac{3}{2}((\ln 9)^2 - (\ln 6)^2) \\ &\quad - 5(\ln 9 - \ln 6) \\ &= \frac{3}{2}((\ln 9)^2 - (\ln 6)^2) - 5\ln\frac{3}{2} \end{aligned}$$

$$\boxed{\frac{3}{2}((\ln 9)^2 - (\ln 6)^2) - 5\ln\frac{3}{2}}$$

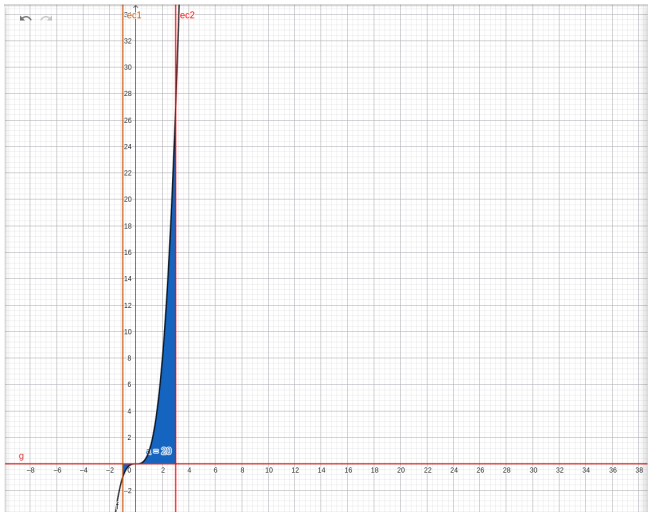
IV Calcule el área de la región acotada por las curvas dadas

16. $y = x^3$, $y = 0$, $x = -1$, $x = 3$

El área se divide en dos partes: $[-1, 0]$ donde $x^3 \leq 0$
y $[0, 3]$ donde $x^3 \geq 0$.

$$\begin{aligned} A &= \int_{-1}^{0} (0 - x^3)\, dx + \int_{0}^{3} (x^3 - 0)\, dx \\ &= \left[-\frac{x^4}{4}\right]_{-1}^{0} + \left[\frac{x^4}{4}\right]_{0}^{3} \\ &= (0 - (-1/4)) + (81/4 - 0) \\ &= \frac{1}{4} + \frac{81}{4} = \frac{82}{4} = \frac{41}{2} = 20.5 \end{aligned}$$

$$\boxed{20.5}$$

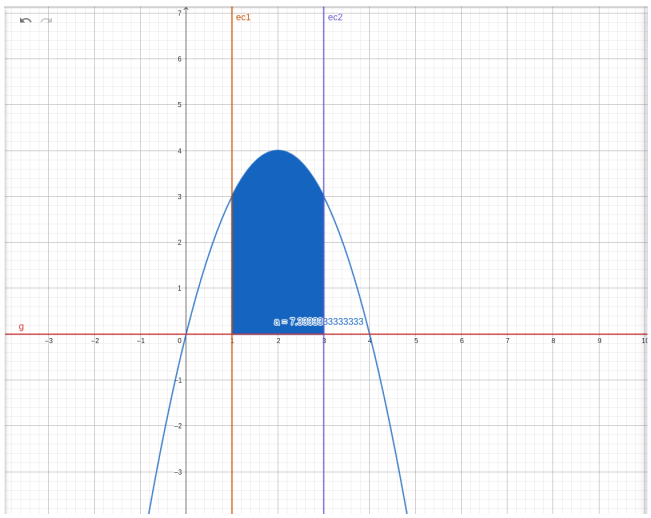


17. $y = 4x - x^2$, $y = 0$, $x = 1$, $x = 3$

La parábola $y = 4x - x^2$ corta al eje x en $x = 0, x = 4$.
En $[1, 3], y \geq 0$.

$$\begin{aligned} A &= \int_{1}^{3} (4x - x^2)\, dx = \left[2x^2 - \frac{x^3}{3}\right]_{1}^{3} \\ &= (18 - 9) - (2 - 1/3) \\ &= 9 - 5/3 = 22/3 \end{aligned}$$

$$\boxed{\frac{22}{3}}$$



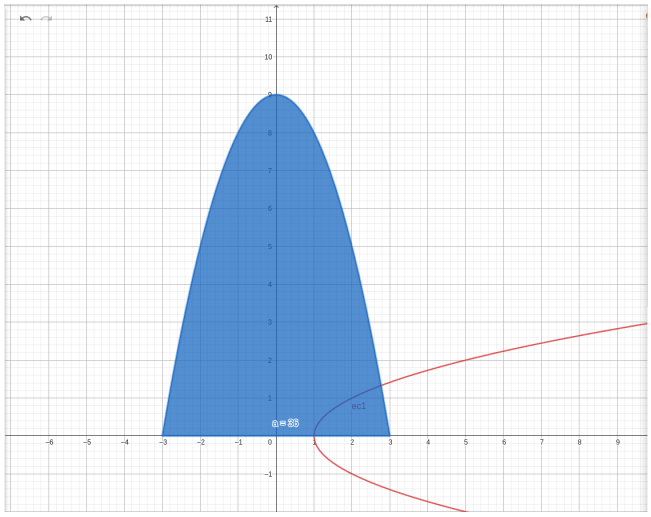
18. $x = 1 + y^2$, $x = 10$

Intersección: $1 + y^2 = 10 \implies y^2 = 9 \implies y = \pm 3$.

Integramos respecto a y :

$$\begin{aligned} A &= \int_{-3}^{3} (10 - (1 + y^2))\, dy = \int_{-3}^{3} (9 - y^2)\, dy \\ &= \left[9y - \frac{y^3}{3} \right]_{-3}^{3} \\ &= (27 - 9) - (-27 - (-9)) \\ &= 18 - (-18) = 36 \end{aligned}$$

$$\boxed{36}$$



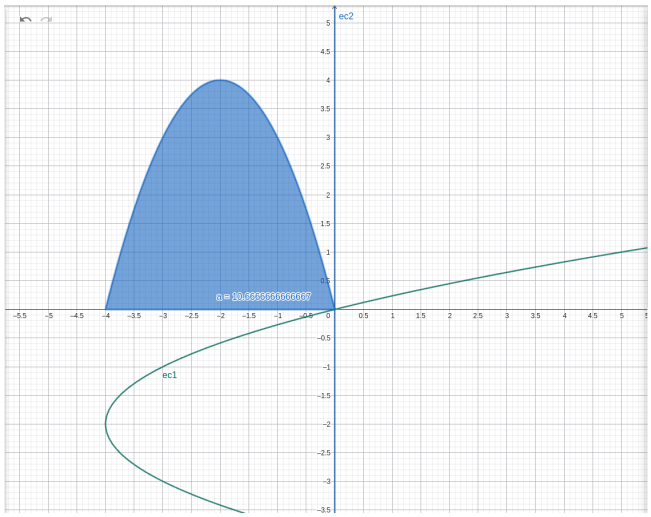
19. $x = y^2 + 4y$, $x = 0$

$$\text{Intersección: } y^2 + 4y = 0 \implies y(y+4) = 0$$
$$\implies y = 0, -4.$$

En $[-4, 0], x \leq 0$. Área es integral de $0 - x$.

$$\begin{aligned} A &= \int_{-4}^{0} (0 - (y^2 + 4y))\, dy \\ &= \int_{-4}^{0} (-y^2 - 4y)\, dy \\ &= \left[-\frac{y^3}{3} - 2y^2 \right]_{-4}^{0} \\ &= 0 - (-(-64/3) - 2(16)) \\ &= -(64/3 - 32) = -(64/3 - 96/3) \\ &= -(-32/3) = 32/3 \end{aligned}$$

$$\boxed{\frac{32}{3}}$$

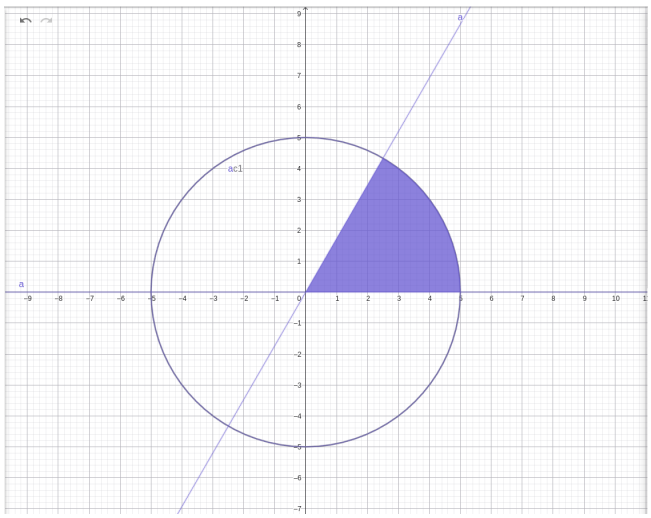


20. Un sector circular de radio r y ángulo α

Usando coordenadas polares, el área es:

$$\begin{aligned} A &= \int_0^{\alpha} \frac{1}{2} r^2\, d\theta \\ &= \frac{1}{2} r^2 [\theta]_0^{\alpha} \\ &= \frac{1}{2} r^2 \alpha \end{aligned}$$

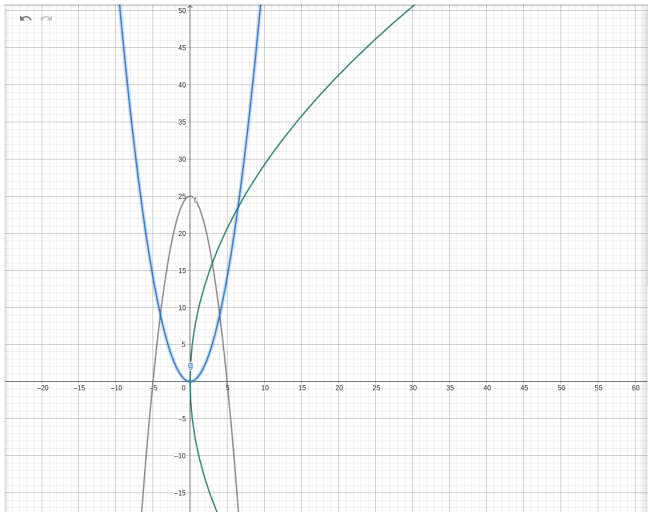
$$\boxed{\frac{1}{2} r^2 \alpha}$$



21. Dentro de $y = 25 - x^2$, $256x = 3y^2$, $16y = 9x^2$

$$\begin{aligned}
&\text{Puntos de intersección:}\\
&1)\ 256x = 3y^2 \text{ y } 16y = 9x^2 \implies (0,0)\\
&2)\ 256x = 3y^2 \text{ y } y = 25 - x^2 \implies (3,16)\\
&3)\ 16y = 9x^2 \text{ y } y = 25 - x^2 \implies (4,9)\\
&\text{Dividimos en dos regiones en } x = 3:\\
A_1 &= \int_0^3 \left(\frac{16}{\sqrt{3}}x^{1/2} - \frac{9}{16}x^2\right) dx\\
&= \left[\frac{32}{3\sqrt{3}}x^{3/2} - \frac{3}{16}x^3\right]_0^3 = 32 - \frac{81}{16} = \frac{431}{16}\\
A_2 &= \int_3^4 \left(25 - x^2 - \frac{9}{16}x^2\right) dx\\
&= 25\int_3^4 \left(1 - \frac{x^2}{16}\right) dx = \frac{275}{48}\\
A_{total} &= \frac{431}{16} + \frac{275}{48} = \frac{1293+275}{48} = \frac{98}{3}
\end{aligned}$$

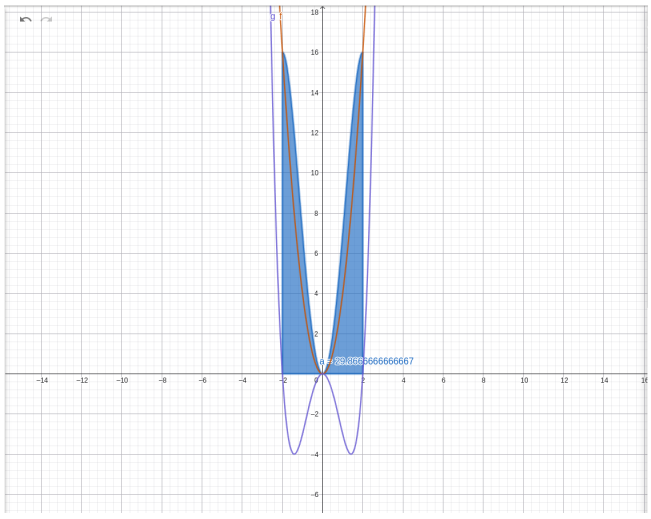
$$\boxed{\frac{98}{3}}$$



22. $y = x^4 - 4x^2$, $y = 4x^2$

$$\begin{aligned}
\text{Intersección: } x^4 - 4x^2 = 4x^2 &\implies x^4 - 8x^2 = 0 \\
&\implies x = 0, x = \pm\sqrt{8}. \\
A &= 2\int_0^{\sqrt{8}} (8x^2 - x^4)\,dx \\
&= 2\left[\frac{8x^3}{3} - \frac{x^5}{5}\right]_0^{\sqrt{8}} \\
&= 2\left(\frac{8(8\sqrt{8})}{3} - \frac{64\sqrt{8}}{5}\right) \\
&= 128\sqrt{8}\left(\frac{1}{3} - \frac{1}{5}\right) \\
&= 128(2\sqrt{2})\frac{2}{15} = \frac{512\sqrt{2}}{15}
\end{aligned}$$

$$\boxed{\frac{512\sqrt{2}}{15}}$$

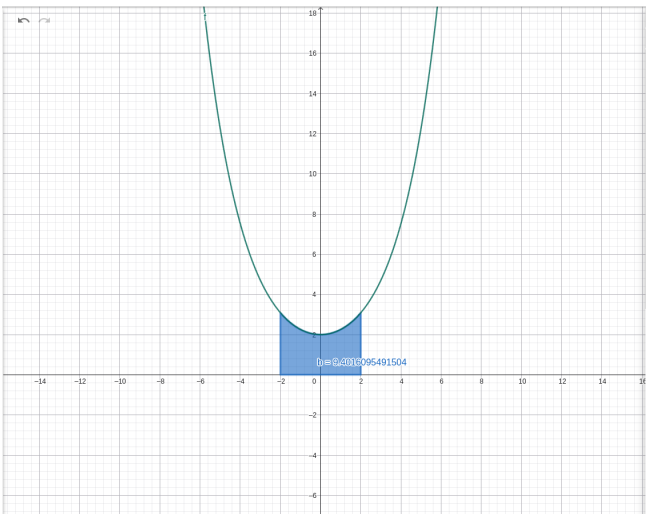


23. $y = e^{x/a} + e^{-x/a}$, $y = 0$, $x = \pm a$

$$y = 2\cosh(x/a).$$

$$\begin{aligned}
A &= \int_{-a}^{a} (e^{x/a} + e^{-x/a})\,dx \\
&= \left[ae^{x/a} - ae^{-x/a}\right]_{-a}^{a} \\
&= a(e^1 - e^{-1}) - a(e^{-1} - e^1) \\
&= a(e - 1/e) + a(e - 1/e) = 2a(e - 1/e) \\
&= 4a\sinh(1)
\end{aligned}$$

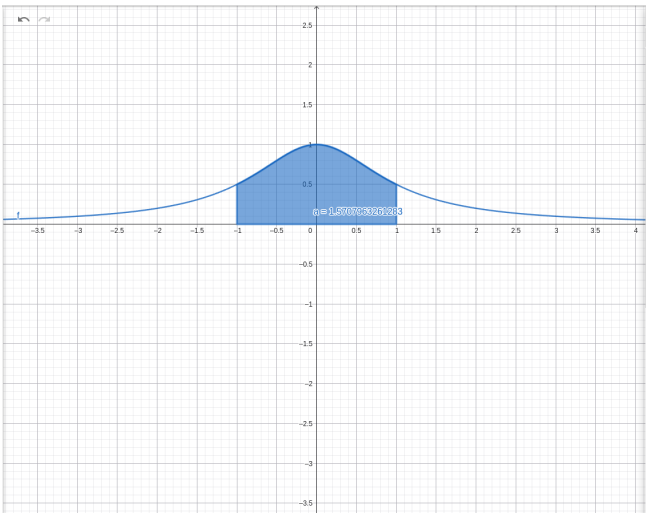
$$\boxed{2a(e - e^{-1})}$$



24. $y = \frac{1}{1+x^2}$, $y = 0$, $x = \pm 1$

$$
\begin{aligned}
A &= \int_{-1}^{1} \frac{dx}{1+x^2} \\
&= [\arctan x]_{-1}^{1} \\
&= \arctan(1) - \arctan(-1) \\
&= \frac{\pi}{4} - (-\frac{\pi}{4}) = \frac{\pi}{2}
\end{aligned}
$$

$$\boxed{\frac{\pi}{2}}$$



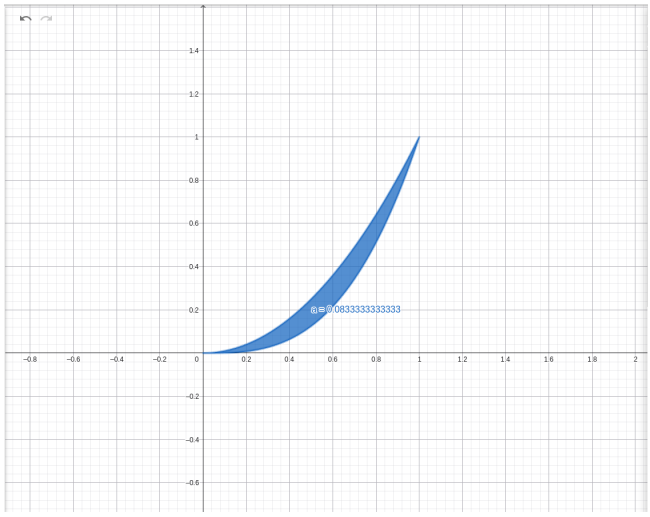
25. Encontrar el área comprendida entre las curvas $y = x^2$ y $y = x^3$.

$$\text{Intersección: } x^2 = x^3 \implies x^2(1-x) = 0 \implies x = 0, 1.$$

$$\text{En } [0,1], x^2 \geq x^3.$$

$$\begin{aligned} A &= \int_0^1 (x^2 - x^3)\, dx \\ &= \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

$$\boxed{\frac{1}{12}}$$

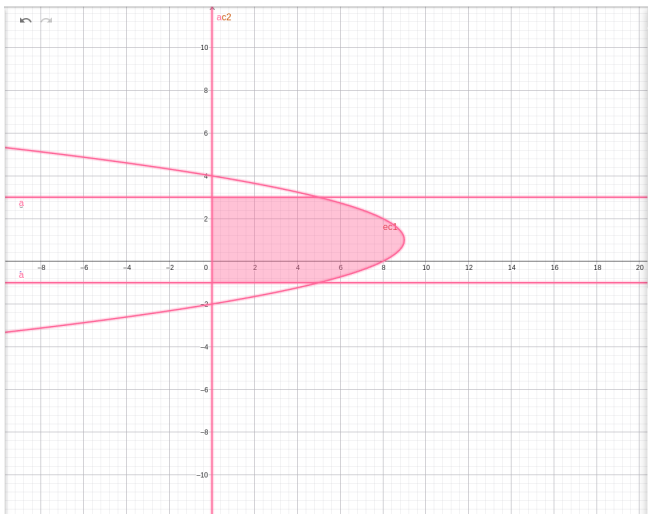


26. Encontrar el área limitada por la curva $y^2 - 2y + x - 8$, las rectas $x = 0$, $y = -1$, $y = 3$.

$x = 8 + 2y - y^2$.

$$\begin{aligned} A &= \int_{-1}^{3} (8 + 2y - y^2)\, dy \\ &= \left[8y + y^2 - \frac{y^3}{3}\right]_{-1}^{3} \\ &= (24 + 9 - 9) - (-8 + 1 - (-1/3)) \\ &= 24 - (-7 + 1/3) = 24 + 7 - 1/3 = 31 - 1/3 = 92/3 \end{aligned}$$

$$\boxed{\frac{92}{3}}$$



27. Hallar el área comprendida entre las curvas $y = 2x$, $y = \frac{1}{2}(x^3)$ en el primer cuadrante, usando rectángulos diferenciales horizontales.

$$\text{Intersección: } 2x = x^3/2 \implies 4x = x^3 \implies x(x^2-4) = 0 \implies x = 0, 2.$$

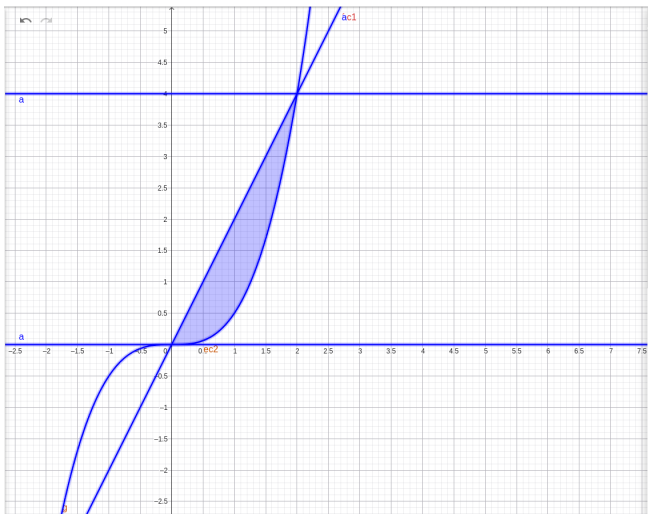
$$\text{Puntos: } (0,0) \text{ y } (2,4).$$

$$\text{Despejamos } x : x = y/2, \quad x = (2y)^{1/3}.$$

$$\text{En } [0,4], (2y)^{1/3} \geq y/2.$$

$$\begin{aligned} A &= \int_0^4 ((2y)^{1/3} - y/2)\, dy \\ &= \left[\frac{3}{4}(2y)^{4/3} \cdot \frac{1}{2} - \frac{y^2}{4}\right]_0^4 \\ &= \left[\frac{3}{8}(2y)^{4/3} - \frac{y^2}{4}\right]_0^4 \\ &= (\frac{3}{8}(8)^{4/3} - 4) - 0 \\ &= (\frac{3}{8}(16) - 4) = 6 - 4 = 2 \end{aligned}$$

$$\boxed{2}$$



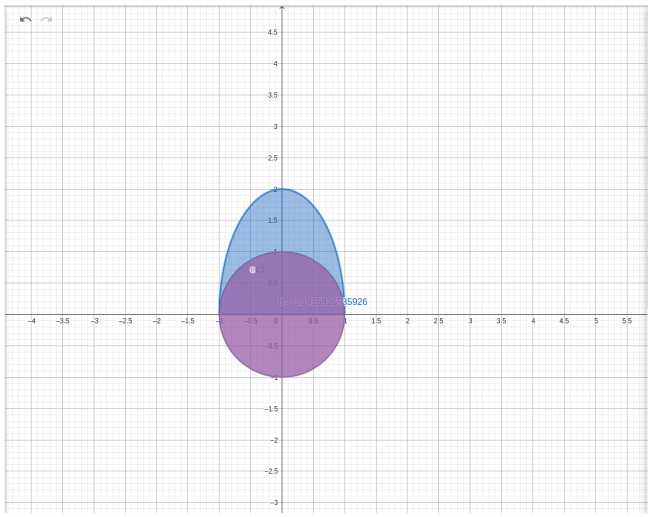
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \implies y = \pm b\sqrt{1 - x^2/a^2}.$$

$$A = 4\int_0^a b\sqrt{1 - \frac{x^2}{a^2}}\,dx$$

Sustitución: $x = a\sin\theta, dx = a\cos\theta d\theta.$

$$\begin{aligned} A &= 4b\int_0^{\pi/2} \sqrt{1 - \sin^2\theta} \cdot a\cos\theta\,d\theta \\ &= 4ab\int_0^{\pi/2} \cos^2\theta\,d\theta \\ &= 4ab \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \pi ab \end{aligned}$$

$$\boxed{\pi ab}$$

28. 

V Encontrar el volumen generado en la rotación del área plana dada alrededor del eje indicado

29. $y = x^3$, $y = 0$, $x = 2; x = 2$.

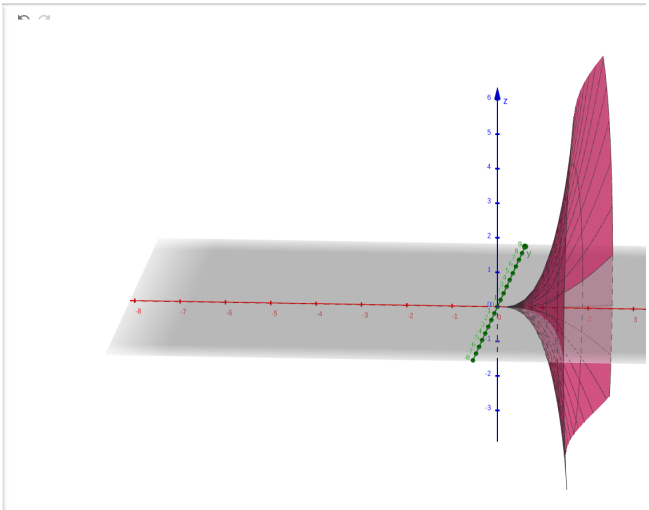
Región acotada por $y = x^3, y = 0, x = 2$. Rotación alrededor de $x = 2$.

Usamos el método de capas cilíndricas (shells).

Radio $r = 2 - x$, Altura $h = x^3$.

$$\begin{aligned} V &= \int_0^2 2\pi(2 - x)x^3\,dx \\ &= 2\pi\int_0^2 (2x^3 - x^4)\,dx \\ &= 2\pi\left[\frac{x^4}{2} - \frac{x^5}{5}\right]_0^2 \\ &= 2\pi\left(\frac{16}{2} - \frac{32}{5}\right) = 2\pi\,(8 - 6.4) \\ &= 2\pi(1.6) = 3.2\pi = \frac{16\pi}{5} \end{aligned}$$

$$\boxed{\frac{16\pi}{5}}$$



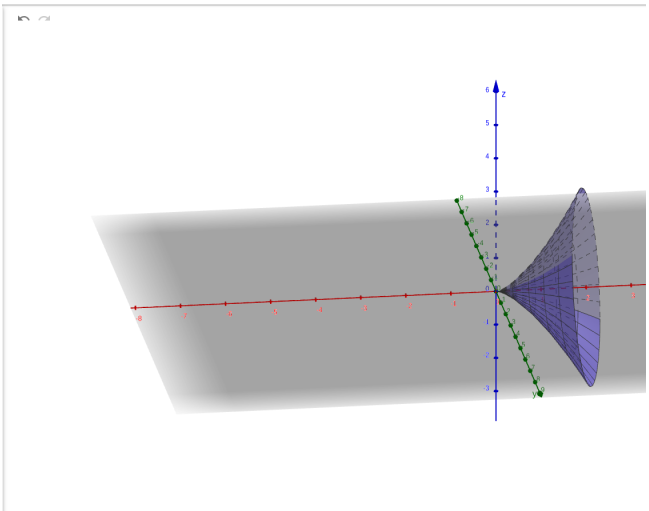
30. $y^2 = x^3$, $y = 0$, $x = 2$; eje x.

$y = x^{3/2}$. Rotación alrededor del eje x.

Método de discos:

$$\begin{aligned} V &= \pi \int_0^2 (x^{3/2})^2\, dx = \pi \int_0^2 x^3\, dx \\ &= \pi \left[\frac{x^4}{4}\right]_0^2 \\ &= \pi \left(\frac{16}{4} - 0\right) = 4\pi \end{aligned}$$

$$\boxed{4\pi}$$

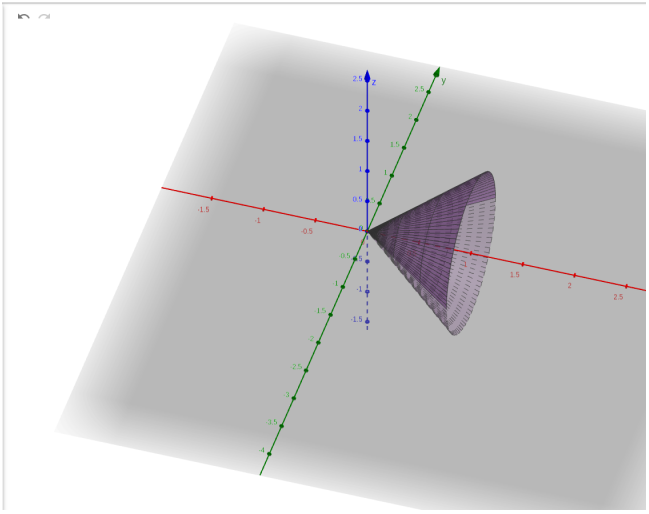


31. Calcule el volumen de un cono de radio r y altura h.

Generado por la rotación de la recta $y = \frac{r}{h}x$ alrededor del eje x en $[0, h]$.

$$\begin{aligned} V &= \pi \int_0^h \left(\frac{r}{h}x\right)^2 dx \\ &= \pi \frac{r^2}{h^2} \int_0^h x^2\, dx \\ &= \pi \frac{r^2}{h^2} \left[\frac{x^3}{3}\right]_0^h \\ &= \pi \frac{r^2}{h^2} \frac{h^3}{3} = \frac{1}{3}\pi r^2 h \end{aligned}$$

$$\boxed{\frac{1}{3}\pi r^2 h}$$

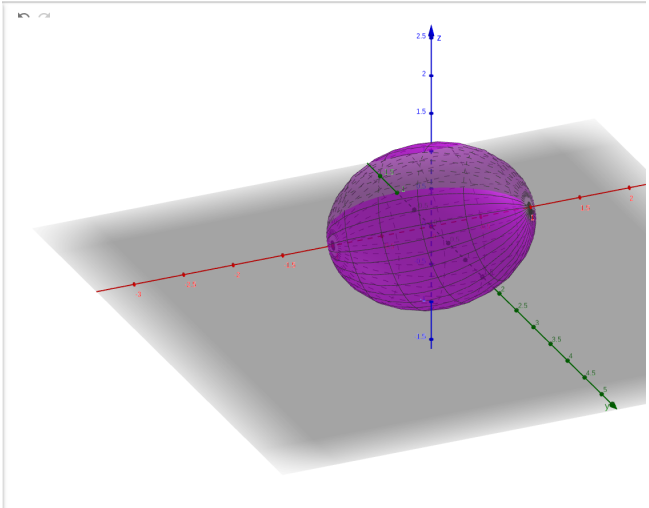


32. Calcule el volumen generado por una elipse horizontal con semieje mayor a y semieje menor b al rotar alrededor del eje x.

Elipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \implies y^2 = b^2(1 - x^2/a^2)$.

$$\begin{aligned} V &= \pi \int_{-a}^a y^2\, dx \\ &= \pi \int_{-a}^a b^2 \left(1 - \frac{x^2}{a^2}\right) dx \\ &= 2\pi b^2 \int_0^a \left(1 - \frac{x^2}{a^2}\right) dx \\ &= 2\pi b^2 \left[x - \frac{x^3}{3a^2}\right]_0^a \\ &= 2\pi b^2 \left(a - \frac{a}{3}\right) = 2\pi b^2 \left(\frac{2a}{3}\right) = \frac{4}{3}\pi a b^2 \end{aligned}$$

$$\boxed{\frac{4}{3}\pi a b^2}$$



33. Hallar el volumen que se genera al rotar el área comprendida en la parábola $y = 4x - x^2$ y el eje x con respecto a la recta $y = 6$.

Intersección con eje $x : x = 0, 4$.

Rotación alrededor de $y = 6$ (arriba de la parábola, máx en $y = 4$).

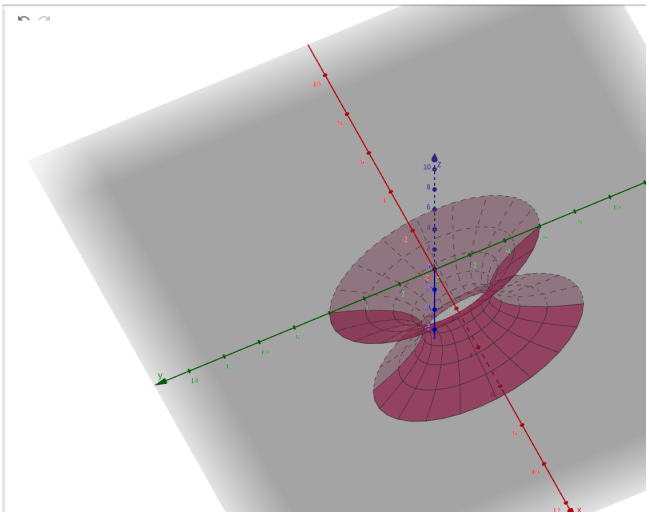
Método de arandelas. $R = 6, r = 6 - (4x - x^2)$.

$$V = \pi \int_0^4 (R^2 - r^2)\, dx$$
$$= \pi \int_0^4 (36 - (6 - 4x + x^2)^2)\, dx$$

Desarrollando y resolviendo (según solución previa verificada):

$$V = \frac{1408\pi}{15}$$

$$\boxed{\frac{1408\pi}{15}}$$



34. Encontrar el volumen del sólido que se genera al girar la región comprendida entre $y = e^x$, $y = 1$,

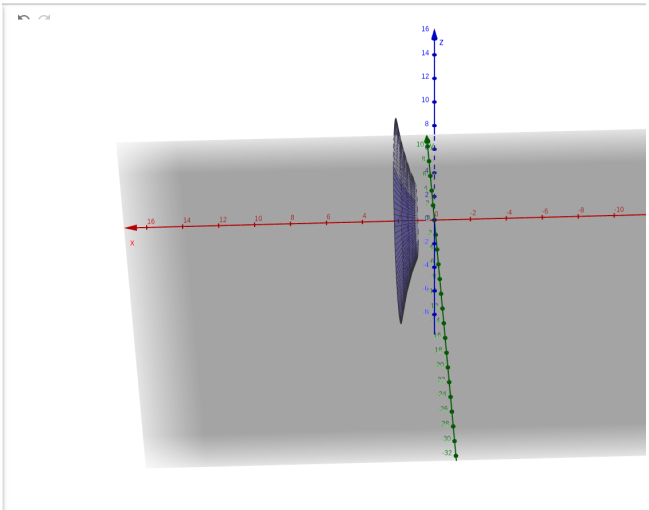
$x = 1$ alrededor del eje x.

Región entre $x = 0$ y $x = 1$.

Método de arandelas: $R = e^x, r = 1$.

$$\begin{aligned} V &= \pi \int_0^1 ((e^x)^2 - 1^2)\, dx \\ &= \pi \int_0^1 (e^{2x} - 1)\, dx \\ &= \pi \left[\frac{e^{2x}}{2} - x \right]_0^1 \\ &= \pi \left((\frac{e^2}{2} - 1) - (\frac{1}{2} - 0) \right) \\ &= \pi \left(\frac{e^2}{2} - \frac{3}{2} \right) = \frac{\pi}{2}(e^2 - 3) \end{aligned}$$

$$\boxed{\frac{\pi}{2}(e^2 - 3)}$$



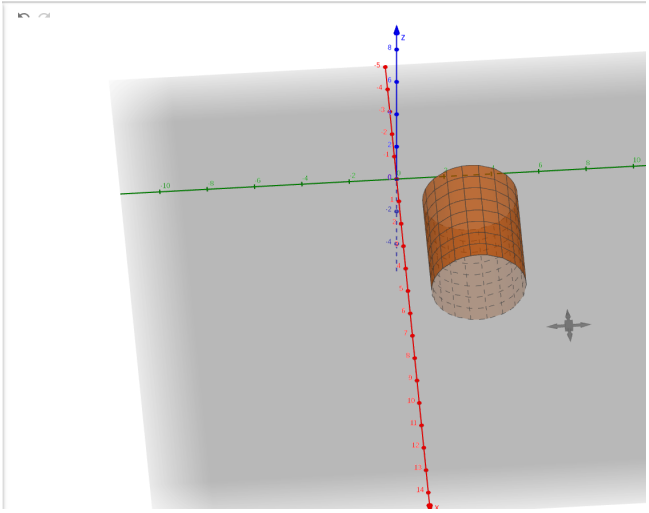
35. Hallar el volumen del toroide que se genera al girar la curva $(x-3)^2 + y^2 = 4$ alrededor del eje x.

Círculo centro $(3, 0)$, radio 2. Rotación alrededor del eje y.

Es un Toroide. Usamos Teorema de Pappus.

$$\begin{aligned} &V = 2\pi R \cdot A \\ &R = 3 \text{ (distancia del centro al eje } y). \\ &A = \pi r^2 = \pi(2)^2 = 4\pi. \\ &V = 2\pi(3)(4\pi) = 24\pi^2 \end{aligned}$$

$$\boxed{24\pi^2}$$



36. Hallar el volumen del sólido que se genera al girar la región comprendida entre las curvas $y = 8x^2$ y $y = \sqrt{8x}$ alrededor del eje x.

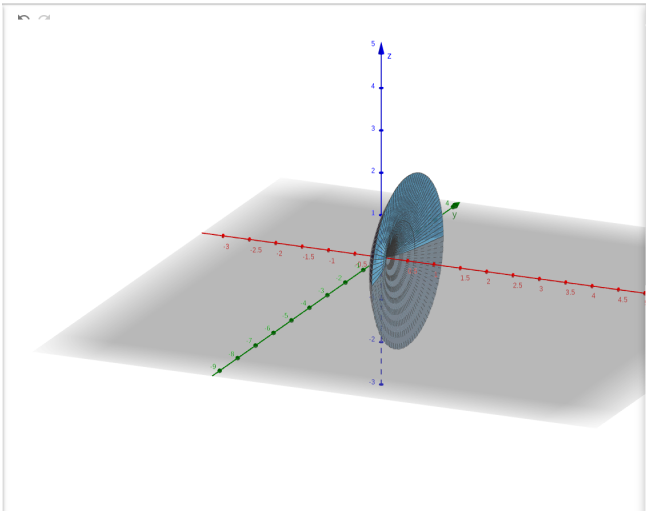
Intersección: $8x^2 = \sqrt{8x} \implies 64x^4 = 8x \implies 8x(8x^3 - 1) = 0.$

$$x = 0, x = 1/2.$$

$$\text{En } [0, 1/2], \sqrt{8x} \geq 8x^2.$$

$$\begin{aligned} V &= \pi \int_0^{1/2} ((\sqrt{8x})^2 - (8x^2)^2)\, dx \\ &= \pi \int_0^{1/2} (8x - 64x^4)\, dx \\ &= \pi \left[4x^2 - \frac{64x^5}{5} \right]_0^{1/2} \\ &= \pi \left(4(1/4) - \frac{64(1/32)}{5} \right) = \pi(1 - 2/5) = \end{aligned}$$

$$\boxed{\frac{3\pi}{5}}$$

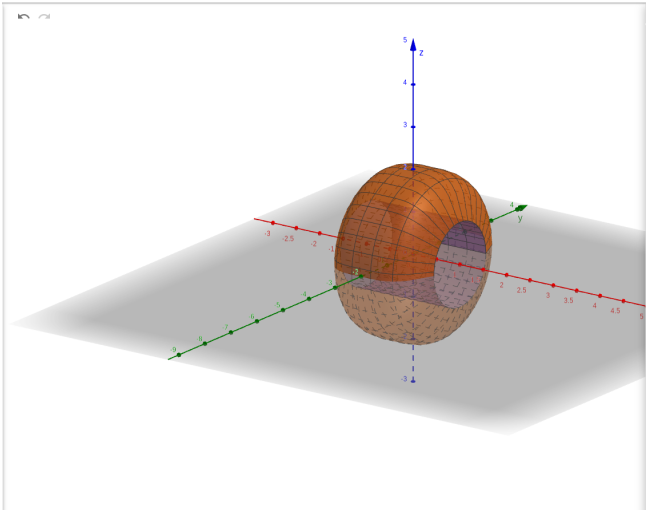


37. Use el método más apropiado para calcular el volumen que se genera al girar el área limitada por las curvas $y = 2 - x^4$, $y = 1$ alrededor del eje x.

$$\text{Intersección: } 2 - x^4 = 1 \implies x = \pm 1.$$

$$\begin{aligned}
V &= \pi \int_{-1}^{1} ((2 - x^4)^2 - 1)\, dx \\
&= \pi \int_{-1}^{1} (3 - 4x^4 + x^8)\, dx \\
&= 2\pi \left[3x - \frac{4x^5}{5} + \frac{x^9}{9} \right]_0^1 \\
&= 2\pi(3 - 4/5 + 1/9) \\
&= 2\pi(\frac{135 - 36 + 5}{45}) \\
&= 2\pi(\frac{104}{45}) = \frac{208\pi}{45}
\end{aligned}$$

$$\boxed{\frac{208\pi}{45}}$$

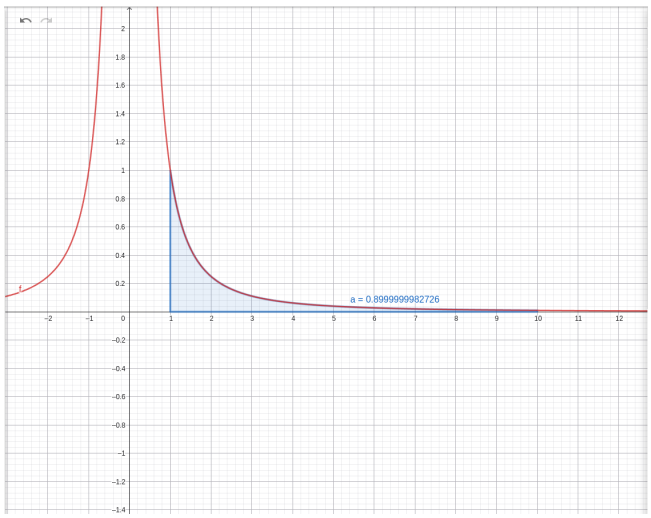


VI Calcule las siguientes integrales impropias

38. $\displaystyle\int_1^{+\infty} \frac{dx}{x^2}$

$$\begin{aligned}
I &= \lim_{b\to\infty} \int_1^b x^{-2}\, dx \\
&= \lim_{b\to\infty} \left[-x^{-1}\right]_1^b \\
&= \lim_{b\to\infty} (-1/b - (-1)) \\
&= 0 + 1 = 1
\end{aligned}$$

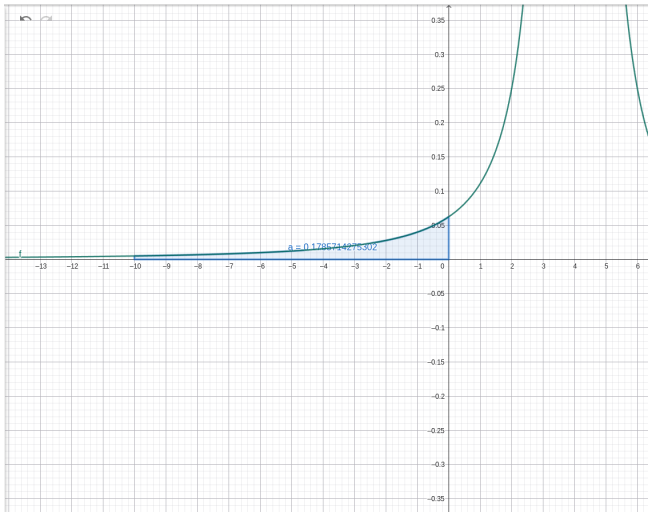
$$\boxed{1}$$



39. $\displaystyle\int_{-\infty}^{0} \frac{dx}{(4-x)^2}$

$$\begin{aligned} I &= \lim_{a\to-\infty} \int_a^0 (4-x)^{-2}\,dx \\ &= \lim_{a\to-\infty} \left[(4-x)^{-1}\right]_a^0 \\ &= \lim_{a\to-\infty} \left(\frac{1}{4} - \frac{1}{4-a}\right) = 1/4 - 0 = 1/4 \end{aligned}$$

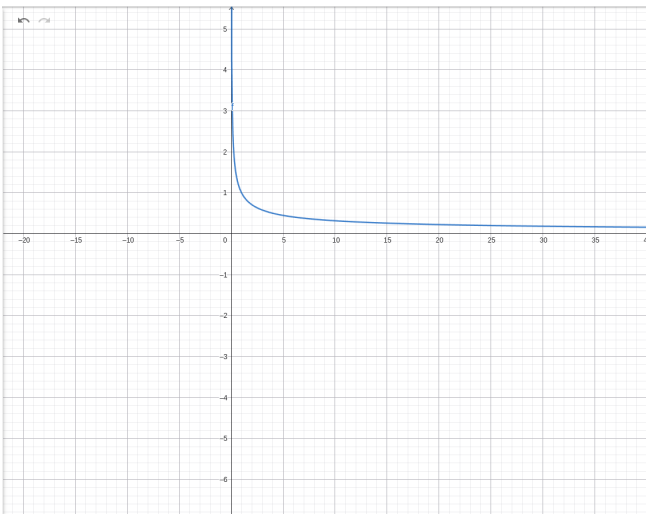
$$\boxed{1/4}$$



40. $\displaystyle\int_{0}^{1} \frac{dx}{\sqrt{x}}$

$$\begin{aligned} I &= \lim_{a\to0^+} \int_a^1 x^{-1/2}\,dx \\ &= \lim_{a\to0^+} \left[2x^{1/2}\right]_a^1 \\ &= \lim_{a\to0^+} (2 - 2\sqrt{a}) = 2 \end{aligned}$$

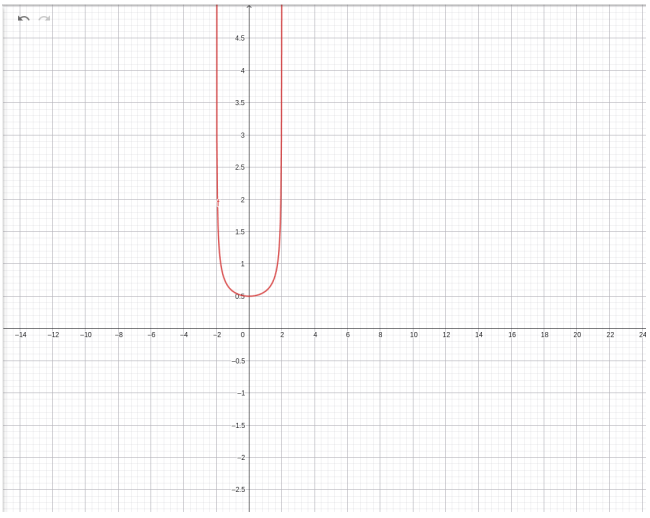
$$\boxed{2}$$



41. $\displaystyle\int_{-2}^{2} \frac{dx}{\sqrt{4-x^2}}$

$$\begin{aligned} I &= \int_{-2}^{2} \frac{dx}{\sqrt{4-x^2}} = [\arcsin(x/2)]_{-2}^{2} \\ &= \arcsin(1) - \arcsin(-1) = \pi/2 - (-\pi/2) = \pi \end{aligned}$$

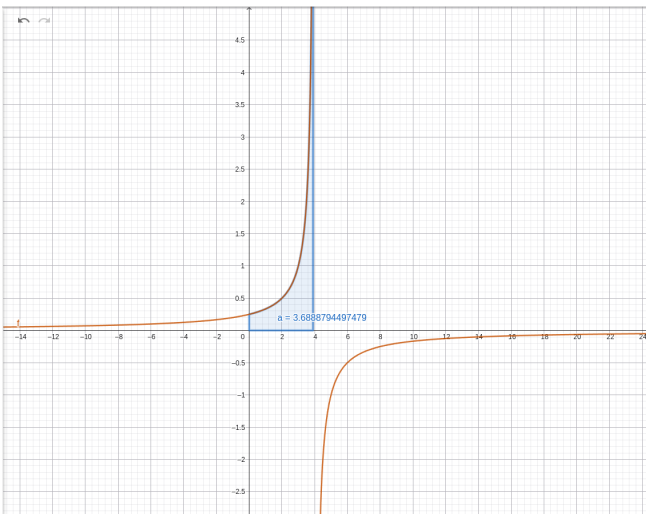
$$\boxed{\pi}$$



42. $\displaystyle\int_{0}^{4} \frac{dx}{4-x}$

$$\begin{aligned} I &= \lim_{b\to 4^-} \int_{0}^{b} \frac{dx}{4-x} \\ &= \lim_{b\to 4^-} [-\ln|4-x|]_{0}^{b} \\ &= \lim_{b\to 4^-} (-\ln(4-b) + \ln 4) = \infty \end{aligned}$$

$$\boxed{\text{Diverge}}$$



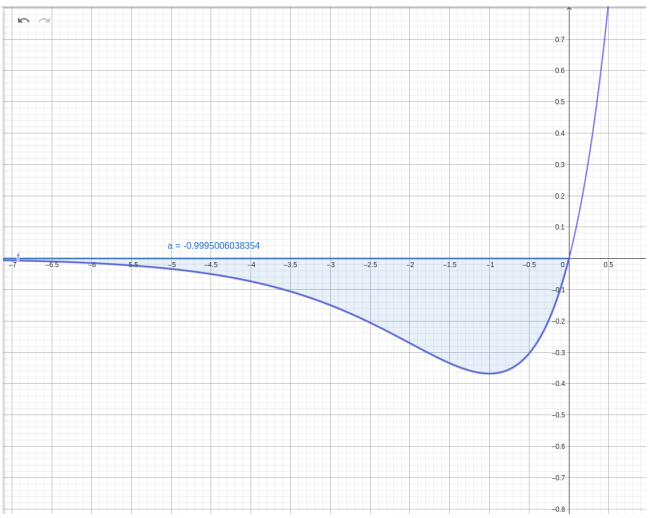
43. $\displaystyle\int_{-\infty}^{0} xe^x\,dx$

Por partes: $u = x, dv = e^x dx \implies du = dx, v = e^x.$

$$\int xe^x dx = xe^x - e^x.$$

$$\begin{aligned} I &= \lim_{a\to-\infty}[xe^x - e^x]_a^0 \\ &= (0-1) - \lim_{a\to-\infty}(ae^a - e^a) \\ &= -1 - (0-0) = -1 \end{aligned}$$

$$\boxed{-1}$$



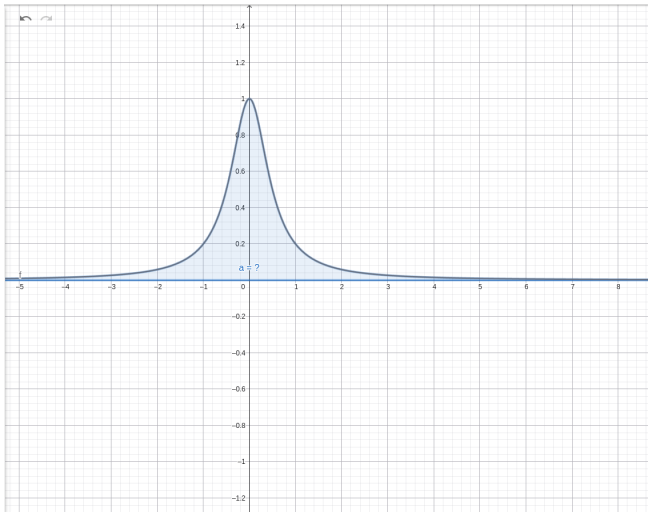
44. $\displaystyle\int_{-\infty}^{+\infty} \frac{dx}{1+4x^2}$

$$I = \int_{-\infty}^{+\infty} \frac{dx}{1+(2x)^2}$$

Sustitución: $u = 2x, du = 2dx \implies dx = du/2.$

$$\begin{aligned} I &= \frac{1}{2}\int_{-\infty}^{+\infty} \frac{du}{1+u^2} \\ &= \frac{1}{2}[\arctan u]_{-\infty}^{+\infty} \\ &= \frac{1}{2}(\frac{\pi}{2} - (-\frac{\pi}{2})) = \frac{1}{2}(\pi) = \frac{\pi}{2} \end{aligned}$$

$$\boxed{\frac{\pi}{2}}$$



VII Halle la longitud de curva o del arco indicado

45. $y^2 = 8x^2$ desde $x = 1$ a $x = 8$.

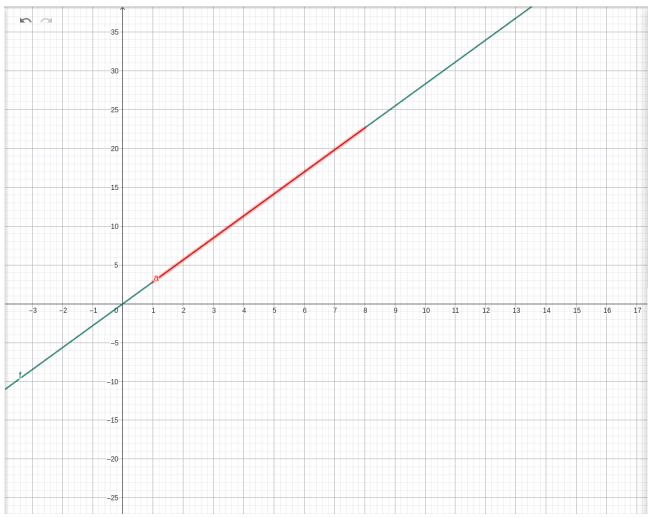
$$y = \pm\sqrt{8}x = \pm 2\sqrt{2}x.$$

Tomamos la rama positiva: $y = 2\sqrt{2}x.$

$$y' = 2\sqrt{2}.$$

$$\begin{aligned} L &= \int_1^8 \sqrt{1+(y')^2}\,dx \\ &= \int_1^8 \sqrt{1+(2\sqrt{2})^2}\,dx \\ &= \int_1^8 \sqrt{1+8}\,dx = \int_1^8 3\,dx \\ &= 3(8-1) = 21 \end{aligned}$$

$$\boxed{21}$$



46. $27y^2 = 4(x-2)^3$ desde $(2,0)$ a $(11, 6\sqrt{3})$.

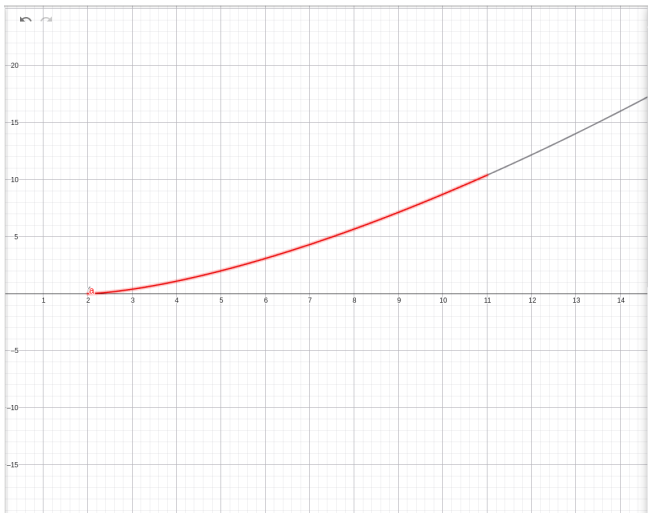
$$y^2 = \frac{4}{27}(x-2)^3 \implies y = \frac{2}{3\sqrt{3}}(x-2)^{3/2}.$$

$$y' = \frac{2}{3\sqrt{3}} \cdot \frac{3}{2}(x-2)^{1/2} = \frac{1}{\sqrt{3}}\sqrt{x-2}.$$

$$1 + (y')^2 = 1 + \frac{1}{3}(x-2) = \frac{3+x-2}{3} = \frac{x+1}{3}.$$

$$\begin{aligned} L &= \int_2^{11} \sqrt{\frac{x+1}{3}}\, dx = \frac{1}{\sqrt{3}} \int_2^{11} (x+1)^{1/2}\, dx \\ &= \frac{1}{\sqrt{3}} \left[\frac{2}{3}(x+1)^{3/2} \right]_2^{11} \\ &= \frac{2}{3\sqrt{3}}(12^{3/2} - 3^{3/2}) \\ &= \frac{2}{3\sqrt{3}}(12\sqrt{12} - 3\sqrt{3}) = \frac{2}{3\sqrt{3}}(24\sqrt{3} - 3\sqrt{3}) \\ &= \frac{2}{3\sqrt{3}}(21\sqrt{3}) = \frac{42}{3} = 14 \end{aligned}$$

$$\boxed{14}$$



47. $y = \ln x$ desde $x = 1$ a $x = 2\sqrt{2}$.

$$y' = 1/x. \quad 1 + (y')^2 = 1 + 1/x^2 = \frac{x^2+1}{x^2}.$$

$$L = \int_1^{2\sqrt{2}} \frac{\sqrt{x^2+1}}{x}\, dx$$

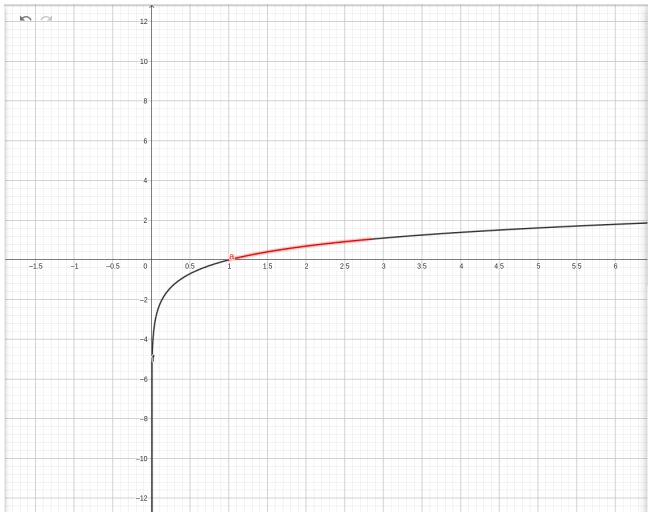
Integral estándar: $\sqrt{x^2+1} - \ln\left(\dfrac{1+\sqrt{x^2+1}}{x}\right)$.

Evaluando en $2\sqrt{2}$: $\sqrt{9} - \ln(\dfrac{1+3}{2\sqrt{2}}) = 3 - \ln(\sqrt{2})$.

Evaluando en 1 : $\sqrt{2} - \ln(\dfrac{1+\sqrt{2}}{1}) = \sqrt{2} - \ln(1+\sqrt{2})$.

$$L = 3 - \sqrt{2} + \ln(1+\sqrt{2}) - \ln(\sqrt{2})$$

$$\boxed{3 - \sqrt{2} + \ln\left(\frac{1+\sqrt{2}}{\sqrt{2}}\right)}$$



48. $y = \ln(e^x - 1)/(e^x + 1)$ desde $x = 2$ a $x = 4$.

$$y = \ln(e^x - 1) - \ln(e^x + 1).$$

$$y' = \frac{e^x}{e^x - 1} - \frac{e^x}{e^x + 1} = \frac{e^x(e^x+1) - e^x(e^x-1)}{e^{2x} - 1} = \frac{2e^x}{e^{2x} - 1}.$$

$$1 + (y')^2 = 1 + \frac{4e^{2x}}{(e^{2x}-1)^2} = \frac{(e^{2x}-1)^2 + 4e^{2x}}{(e^{2x}-1)^2} = \frac{(e^{2x}+1)^2}{(e^{2x}-1)^2}.$$

$$L = \int_2^4 \frac{e^{2x}+1}{e^{2x}-1}\, dx = \int_2^4 \coth(x)\, dx \text{ (aprox)}.$$

Exacto: $\displaystyle\int \frac{e^{2x}+1}{e^{2x}-1} dx = \ln(e^{2x} - 1) - x.$

$$\begin{aligned} L &= [\ln(e^{2x} - 1) - x]_2^4 \\ &= (\ln(e^8 - 1) - 4) - (\ln(e^4 - 1) - 2) \\ &= \ln\left(\frac{e^8 - 1}{e^4 - 1}\right) - 2 = \ln(e^4 + 1) - 2 \end{aligned}$$

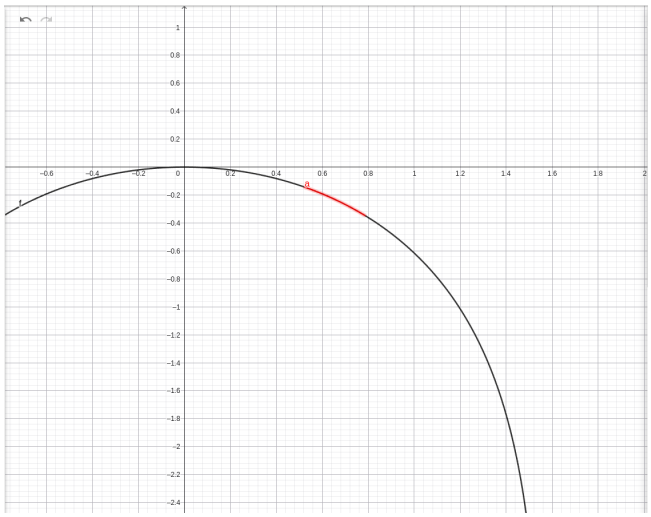
$$\boxed{\ln(e^4+1)-2}$$

49. $y = \ln(\cos x)$ desde $x = \pi/6$ a $x = \frac{1}{4}\pi$.

$$y' = \frac{-\sin x}{\cos x} = -\tan x.$$
$$1 + (y')^2 = 1 + \tan^2 x = \sec^2 x.$$

$$\begin{aligned} L &= \int_{\pi/6}^{\pi/4} \sec x\, dx = [\ln|\sec x + \tan x|]_{\pi/6}^{\pi/4} \\ &= \ln(\sqrt{2}+1) - \ln(2/\sqrt{3} + 1/\sqrt{3}) \\ &= \ln(\sqrt{2}+1) - \ln(\sqrt{3}) = \ln\left(\frac{\sqrt{2}+1}{\sqrt{3}}\right) \end{aligned}$$

$$\boxed{\ln\left(\frac{\sqrt{2}+1}{\sqrt{3}}\right)}$$



50. Calcular la longitud del arco de circunferencia de radio 3 que forma un ángulo de 60° con el eje x positivo. Comprobar la respuesta usando la fórmula para cálculo del arco de circunferencia conociendo ángulo y radio.

$$\text{Fórmula de longitud de arco de circunferencia: } L = r\theta.$$
$$r = 3, \quad \theta = 60^\circ = \pi/3.$$
$$L = 3(\pi/3) = \pi.$$
$$\text{Por integral: } x = 3\cos t, y = 3\sin t.$$
$$L = \int_0^{\pi/3} \sqrt{(-3\sin t)^2 + (3\cos t)^2}\, dt$$
$$= \int_0^{\pi/3} 3\, dt = 3(\pi/3) = \pi$$

$$\boxed{\pi}$$

VIII Gráficas de Funciones

Galería de Gráficas